M^{2}SegMamba: Mamba-Based Incomplete Multimodal Learning for Brain Tumor Segmentation With Few Samples

Xinyue Zhang, Ali Bahri, Christian Desrosiers, Hui Liu, and Fangxun Bao

***Abstract*—The accurate segmentation of brain tumors plays an important role in clinical diagnosis and treatment. Multimodal magnetic resonance imaging (MRI) can provide rich and complementary information for accurate brain tumor segmentation. However, the common problems of incomplete modalities and small samples in clinical practice seriously affect the performance of multimodal segmentation. In this work, we design a new framework, named M^{2}SegMamba, using Mamba and Masked Autoencoder networks for both supervised and self-supervised learning, aimed at handling small sample brain tumor segmentation under various incomplete multimodality settings. We construct a masking strategy suitable for multimodal brain tumors to precisely extract image features, which serves as the foundation for image segmentation. By fully leveraging the capabilities of the Mamba network, we design a multi-traversal method to facilitate the interaction between inter-modal and cross-modal image features. Meanwhile, the introduction of TSmamba in skipping connections efficiently integrates multimodal features. Auxiliary regularizers are introduced in both the encoder and decoder to further enhance the model's robustness to incomplete modalities. We conducted experiments on the BraTS 2018 and BraTS 2020 datasets, and the results demonstrate that our method outperforms state-of-the-art brain tumor segmentation methods on most subsets of missing modalities.**

***Index Terms*—Brain tumor segmentation, hetero-modal learning, state-space model, masked image modeling.**

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The code is available at: https://github.com/xin131419/M2SegMamba.git.

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I. Introduction

SEGMENTING brain tumors like gliomas in multimodal MRI, which involves distinguishing cancerous regions—such as the whole tumor, core tumor, and enhanced tumor—from healthy brain tissue, is a vital step for diagnosis and treatment. However, the inherent heterogeneity of tumors, in terms of appearance, shape, and varying intensity profiles across multimodal MRI scans (e.g., Fig. 1), poses a significant challenge for this task [1]. Furthermore, due to differences in scanning protocols, and patient-related limitations, it is often impossible to obtain a complete set of MRI sequences for segmentation. This has prompted researchers to develop *hetero-modal* learning methods [2], [3], [4], [5], [6] which can handle cases where any subset of modalities is not available at inference.

In recent years, Vision Transformers (ViTs) have gained increasing popularity for their ability to model global interactions in images and seamlessly integrate different input modalities. Unlike convolutional neural networks (CNN) which extract local features using convolution filters, ViTs represent images as a set of patch tokens and combine information across tokens via their self-attention mechanism [7]. Several studies in medical imaging have explored the potential of ViTs for multimodal brain tumor segmentation. Zhang et al. [3] proposed a hybrid architecture called *mmFormer*, where available modalities are first processed separately by a sequence of CNN and ViT blocks, and then combined with a modality-correlated encoder based on self-attention. Li et al. [8] trained a CNN with deformable convolutions and cross-attention to extract and combine features from a potentially incomplete set of modalities. The proposed model, named *SRMNet*, comprises two jointly-trained decoders, one predicting the segmentation mask of tumor regions, and the other reconstructing input images. In their *M^{2}FTrans* method, Shi et al. [9] enhanced cross-modality interaction and robustness to missing modalities by introducing learnable fusion tokens and computing masked self-attention.

Although the self-attention mechanism in ViTs enables modeling global dependencies in images, its quadratic complexity with respect to the number of tokens restricts its scalability for large inputs. This challenge is especially critical in multimodal segmentation where processing multiple 3D scans is required. State Space Model (SSM) [10], [11], [12] have recently emerged as an efficient alternative to Transformers, offering a unique capability to model global relationships with linear complexity.

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The Mamba model [13], in particular, has demonstrated significant promise across various visual understanding tasks [14], [15], [16], [17]. Although recent studies have explored the effectiveness of Mamba for medical image segmentation [18], [19], [20], [21], its application to hetero-modal scenarios with missing modalities remains unexplored.

In this paper, we bridge this gap by introducing the first Mamba-based approach for multimodal brain segmentation, capable of handling missing modalities. Our method harnesses the efficiency of SSMs like Mamba to integrate information from multiple 3D scans while minimizing computational and memory overhead. Furthermore, to address the challenge of obtaining expert annotations for medical image segmentation, we propose a self-supervised learning strategy, which significantly reduces the need for labeled data during training. Using this strategy, our method achieves state-of-art performance across scenarios with varying degrees of labeled data scarcity.

The main contributions of our work are listed as follows:

1) We present M^{2}SegMamba, the first Mamba-based method for 3D multimodal image segmentation with missing modalities. Our method effectively captures global interactions within entire images and across different modalities, while minimizing computational and memory requirements.
2) We introduce a novel self-supervised pre-training strategy that utilizes image masking to enhance the robustness of our method, particularly when only a limited number of labeled samples are available. To the best of our knowledge, this is the first study to investigate this strategy within a Mamba-based segmentation model.
3) We assess our M^{2}SegMamba method through comprehensive experiments across a diverse set of scenarios, varying in available modalities and amounts of labeled data. Our findings highlight the robustness and superior performance of M^{2}SegMamba in comparison to existing approaches.

II. Related Work

A. Multimodal Brain Tumor Segmentation

The methods proposed for this fundamental task in medical image analysis can be broadly categorized into two types: those that require the complete set of modalities and those capable of handling missing modalities. Methods in the first category, typically based on CNNs [22], [23] or ViTs [24], [25], [26], adhere to the conventional image segmentation pipeline, maintaining the same input during both training and inference. Conversely, the second category of approaches tackles the challenge of hetero-modal learning, where any subset of modalities may be missing during training or inference. A solution to this problem uses knowledge distillation to transfer the knowledge of a model trained with the full set of modalities to one that takes a subset of these modalities as input [6], [27], [28]. However, this approach can be costly to implement in practice, as a separate model is needed for each possible subset of missing modalities. Another strategy employs a generative model to synthesize the missing modalities. For example, Huang et al. [29] proposed a context-aware generative adversarial network (CoCa-GAN), based on an encoder-decoder architecture, where the decoder generates the missing target modality. One limitation of this strategy is that it requires training a separate generative network for each modality and is highly sensitive to the quality of the generated images.

Rather than learning modality-specific networks, another group of methods focuses on developing a unified model capable of directly handling missing modalities. For example, the HeMIS model by Havaei et al. [2] integrates feature embeddings from an arbitrary set of modalities by calculating their mean and variance. Ding et al. [5] proposed the Region-aware Fusion Network (RFNet), which learns a probability map to segment multimodal features into distinct regions and applies attention weights within each region to dynamically adjust the contributions of different image modalities.

B. State Space Models

State Space Models (SSMs) [30] have gained increasing attention as sequence-to-sequence models for their ability to efficiently capture long-range dependencies while supporting parallel training. Compared to the self-attention mechanism in Transformers, which computes an attention weight for each pair of input tokens, SSMs model the interdependence between tokens through latent state variables that are updated sequentially. Mamba [13], a widely used SSM, enhanced the structured state-space sequence (S4) model [31] by incorporating a data-dependent selection mechanism. Adapting this model to vision tasks, VMamba [32] designed a Cross-Scan Module (CSM) that extends 1D selective scanning to the domain of 2D images. Similarly, Vision Mamba [33] introduced Vim blocks that combine positional embeddings with bidirectional sequence modeling. CU-Mamba [34] proposed integrating a dual state space model framework into U-Net structure. It also adopted a Spatial SSM module for global context encoding as well as a Channel SSM component to retain channel-related features.

Recent studies have introduced SSM-based models for medical image segmentation. Xing et al. [18] proposed SegMamba, an innovative 3D image segmentation model that efficiently captures long-range dependencies across whole-volume features at multiple scales. Ma et al. [35] developed a hybrid CNN-SSM block, which combines the local feature extraction power of convolutional layers with the long-range dependency modeling capabilities of SSMs, enabling its application to both 2D and 3D biomedical image segmentation. Our work builds upon these previous studies by introducing the first Mamba-based approach for 3D multimodal image segmentation in the presence of missing modalities.

C. Masked Image Modeling

Masked image modeling (MIM) is a self-supervised learning technique in computer vision that trains models by masking parts of the input—such as pixels, patches, or latent representations—and using the remaining visible context to predict the masked regions and learn meaningful representations [36]. Research in MIM primarily explores two approaches: comparing latent

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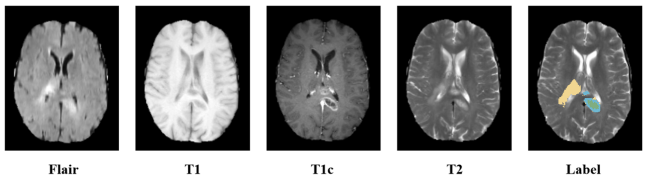


Fig. 1. Four modalities of brain tumor images and corresponding segmentation labels. In the segmentation results, different colors represent different brain tumor regions.

representations and reconstructing the masked signals. The first approach involves comparing the original or weakly augmented input image with a version where parts of the image are masked and strongly augmented, enabling the model to learn robust features [37], [38]. The second approach revolves around masking some information at any stage during the forward pass of the model and using the decoder to reconstruct the missing data [39], [40], [41]. BEiT [39] randomly masks some image patches and feeds them into the Transformer to recover the visual tokens using a well-trained tokenizer. He et al. [40] proposed an asymmetric encoder-decoder architecture that takes only a few visible tokens as input. This design accelerates the training process of MIM while keeping the computational cost relatively low. M^{3}AE [42] follows this reconstruction-based strategy and enhances performance on multi-modal medical segmentation tasks by introducing an efficient self-distillation framework.

In this work, we utilize Masked Image Modeling (MIM) to pre-train a network for brain tumor segmentation with incomplete multimodal MRI data. By reconstructing masked segments of the network's latent features, our method efficiently learns a more robust representation of multimodal images.

III. Method

Our Mamba-based approach for multimodal brain tumor segmentation with missing modalities comprises two key steps: *1*) a pre-training phase focused on masked feature reconstruction (Fig. 2), and *2*) a fine-tuning phase for optimizing the segmentation task (Fig. 4). Before outlining these two steps, we first provide some background information on Mamba, which serves as the context modeling component in our method.

A. Preliminaries

Mamba derives from SSMs [30] like S4 that implement a Linear Time-Invariant (LTI) system mapping a one-dimensional input signal $x(t) \in \mathbb{R}$ to an N-dimensional latent state $h(t) \in \mathbb{R}^N$, and then projecting it onto a one-dimensional output signal $y(t) \in \mathbb{R}$. This system can be expressed using a the following linear ordinary differential equation:

$$\begin{aligned} h'(t) &= Ah(t) + Bx(t) \\ y(t) &= Ch(t) \end{aligned} \tag{1}$$

where $A \in \mathbb{R}^{N\times N}$ is the state transition matrix, $B \in \mathbb{R}^{N\times 1}$ is the input-to-state matrix, and $C \in \mathbb{R}^{1\times N}$ is the state-to-output matrix.

To be compatible with deep learning frameworks, SSMs must undergo a discretization process that divides continuous time based on time scale parameter $\Delta \in \mathbb{R}$. A popular strategy for this process is the Zero-Order Hold (ZOH) rule, computing the discretized versions of A and B as

$$\begin{aligned} \overline{A} &= \exp(\Delta A), \\ \overline{B} &= (\Delta A)^{-1}\left(\exp(\Delta A) - I\right)\Delta B. \end{aligned} \tag{2}$$

This yields the following discretized system:

$$\begin{aligned} h_t &= \overline{A}h_{t-1} + \overline{B}x_t, \\ y_t &= Ch_t. \end{aligned} \tag{3}$$

Since $\overline{A}$ and $\overline{B}$ are linear operators, the recurrent formulation of (3) can be unrolled into single convolution with kernel $\overline{K} \in \mathbb{R}^L$, where L is the input length, enabling parallel computations:

$$y = x \odot \overline{K}, \quad \overline{K} = \left(C\overline{B}, C\overline{AB}, \ldots, C\overline{A}^{L-1}\overline{B}\right). \tag{4}$$

Although efficient, time-invariant models lack the capacity to selectively emphasize or ignore specific parts of the input. To enhance content-aware reasoning, Mamba [13] introduced a time-variant model called Selective SSM, in which the parameters B, C, and Δ dynamically adjust based on the input. It also introduced several strategies, such as parallel scanning and kernel fusion, to enhance computational efficiency.

B. Masked Image Reconstruction Pre-Training

As shown in Fig. 2, we utilize a 3D autoencoder architecture in the pre-training stage to capture both local and contextual features through multimodal image reconstruction. Unlike standard autoencoders, which consist of a symmetric encoder and decoder, our architecture has a dedicated encoder for each modality. These modality-specific encoders integrate convolutional and Mamba blocks to effectively capture both local and global interactions within and across multimodal images, ensuring robust performance even in the presence of missing modalities.

The features extracted by encoders are combined using our proposed multimodal traversal strategy, which processes 3D patches in different sequential orders accounting for both spatial position and modality. This enables the model to learn intra- and inter-modality dependencies between patches. For efficiency, a single convolutional decoder is employed to reconstruct the images across all modalities, including missing ones, from the outputs of multimodal traversals. Finally, to capture robust features that encode structural information across multiple modalities, we employ a multimodal masking strategy that randomly masks patch features both within and across different image modalities. The following subsections provide a detailed explanation of these components of our method.

1) Modality-Specific Image Encoder: As shown in Fig. 2, we employ a hybrid architecture for the modality-specific encoders, integrating a 3D convolutional head to extract local context features from 3D images and a Mamba tail to capture global dependencies across 3D patches.

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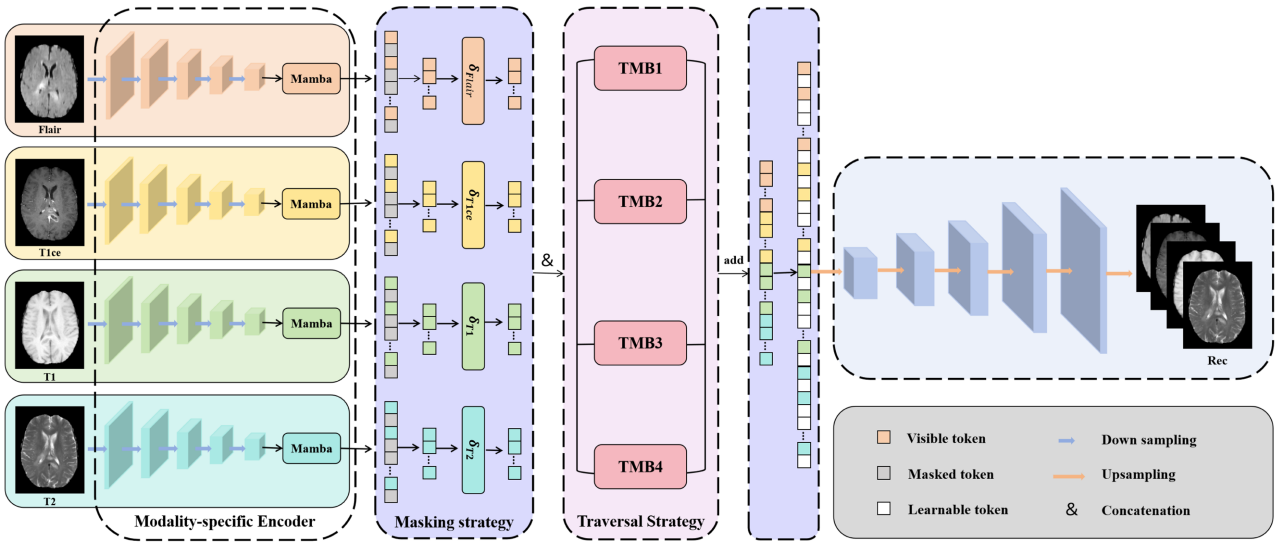


Fig. 2. The overall framework of the 3D multimodal pre-training mask reconstruction network for brain tumors. Each modality has a dedicated specific encoder. After the masking strategy and multiple traversals, the fused features are input into the feature sharing decoder to activate its reconstruction capability.

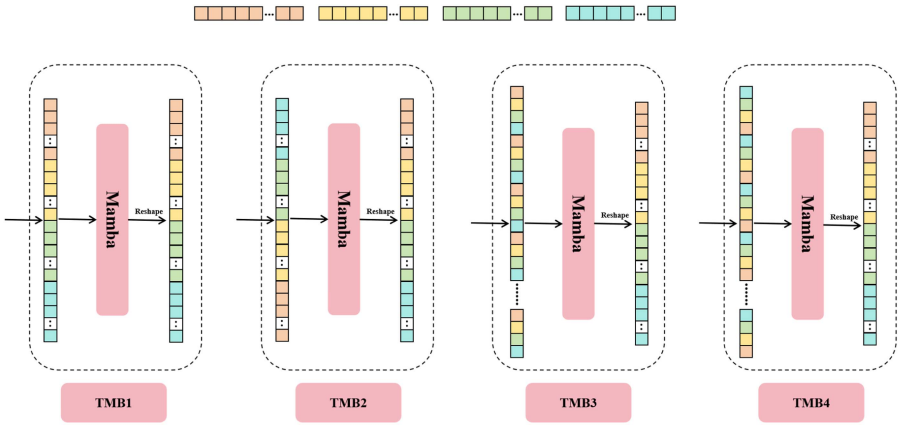


Fig. 3. Multi-traversal methods across modalities. Each color represents a different modality.

Our convolutional head, inspired by the 3D U-Net encoder [43], consists of multiple stacked convolutional blocks, forming a five-stage encoder. Each stage consists of two instance-normalized $3 \times 3 \times 3$ convolutional layers, followed by a ReLU activation. To transition between stages, we employ convolutional layers with a stride of 2, which reduces the feature resolution while proportionally increasing the number of channels. Let $\mathcal{M} = \{\text{Flair}, \text{T1}, \text{T1c}, \text{T2}\}$ be the complete set of modalities. Given a 3D image $X_m \in \mathbb{R}^{H \times W \times D}$ of modality $m \in \mathcal{M}$, having a width W, height H and depth D, we use convolutional head $E_m^{\text{head}}(\cdot)$ to extract local features as

$$F_m^{\text{head}} = E_m^{\text{head}}(X_m) \in \mathbb{R}^{C \times H' \times W' \times D'}. \tag{5}$$

Convolutional networks are inherently limited by their localized nature, which hampers their ability to effectively capture long-range dependencies. To address this challenge, we utilize Mamba blocks, which are designed to model long-range dependencies across features from different modalities. Toward this goal, we first flatten the features of each modality F_m to a shape of (C, L), where $L = H' \times W' \times D'$. We then use these flattened features as input to the modality-specific Mamba encoder tail $E_m^{\text{tail}}(\cdot)$ to obtain the final encoder features for modality m:

$$\hat{F}_m = E_m^{\text{tail}}(F_m^{\text{head}}) \in \mathbb{R}^{C \times L} \tag{6}$$

The function $E_m^{\text{tail}}(\cdot)$ is obtained from (1) to ensure global modeling capability while improving computational efficiency. Feeding the flattened modality features into Mamba not only reduces computational overhead but also enables a more efficient capture of long-range dependencies in the entire image, making it more suitable for large-scale 3D medical image multimodal feature fusion.

2) Multimodal Masking Strategy: Inspired by Masked Image Modeling (MIM) [36], we mask the latent features of randomly selected tokens across different modalities, and train the decoder to reconstruct the corresponding patches. This approach

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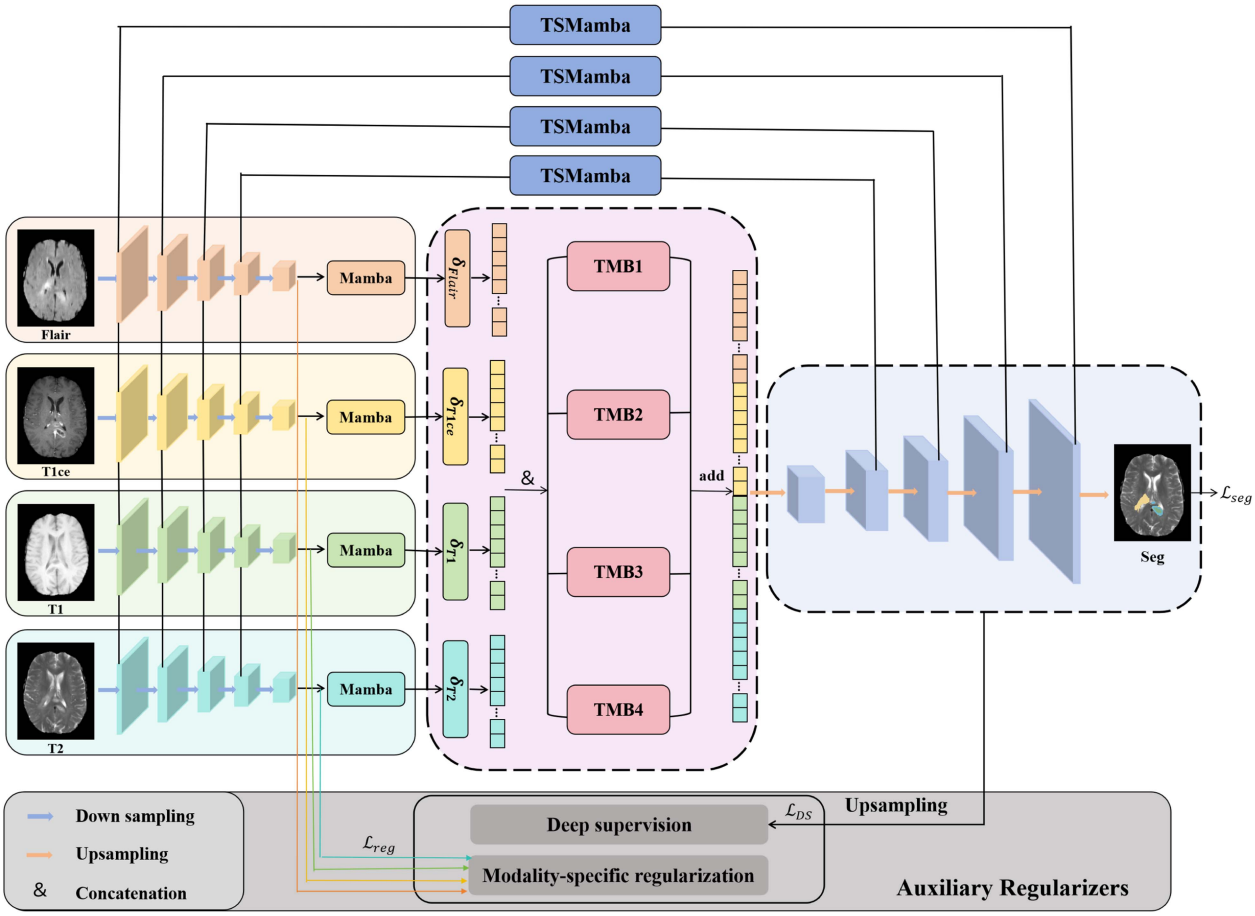


Fig. 4. The fine-tuning framework designed for brain tumor segmentation.

encourages the encoder to learn robust features that capture both intra- and inter-modality dependencies, effectively distinguishing between healthy brain tissue and tumors—enhancing performance in the downstream segmentation task.

Our multimodal masking strategy differs from the traditional approach which randomly selects tokens with a mask ratio of γ and then replaces these tokens with an empty or learnable [masked] token [41]. This standard approach is ineffective in our case because information across modalities is highly correlated (see Fig. 1). As a result, a masked patch in one modality can be easily reconstructed using the corresponding patches from other modalities. To alleviate this problem, we propose a *cross-modal* masking strategy which simultaneously masks the same token (i.e., corresponding to the same patch position) across all modalities.

Additionally, to better align the pre-training step with the downstream application where entire modalities may be missing, we also apply a *modality-wise* masking strategy where *all* tokens of some modalities are masked. Let $\delta_m \in \{0, 1\}$ be an indicator variable such that $\delta_m = 0$ if modality m is masked, else $\delta_m = 1$. The masked features are obtained as

$$F_m = \delta_m \cdot \text{Mask}(\hat{F}_m, M_\gamma),\ m \in \mathcal{M} \tag{7}$$

where $M_\gamma \in \{0, 1\}^L$ is the cross-modal mask common for all modalities, and Mask$(\cdot)$ is a function replacing masked tokens by learnable tokens.

The resulting masked features are fed to the Multimodal Traversal Strategy part of the network for feature fusion. Finally, the decoder receives all tokens—including the learnable masked tokens and encoded visible patches from all modalities—to reconstruct the multimodal image.

3) Multimodal Traversal Strategy: The original Mamba block, designed for processing 1D sequences, lacks the spatial awareness needed for vision tasks. To address this, extensions for image processing [32], [33] introduce a 2D traversal method, such as scanning image patches row-by-row or column-by-column in both forward and reverse directions. Our multimodal traversal strategy, depicted in Fig. 3, further extends this approach by additionally traversing token across different modalities.

Denote as F_m^i the features of the i-th token for modality m, with $i \in \{1, \ldots, L\}$ and $m \in \mathcal{M}$. We traverse tokens following four different scanning routes:

- **TMB1**: Tokens of the four modalities are concatenated directly:

$$\begin{aligned} F_{\text{TMB1}} = [&F^1_{\text{Flair}},\ F^2_{\text{Flair}},\ \ldots,\ F^L_{\text{Flair}},\\ &F^1_{\text{T1c}},\ F^2_{\text{T1c}},\ \ldots,\ F^L_{\text{T1c}},\\ &F^1_{\text{T1}},\ F^2_{\text{T1}},\ \ldots,\ F^L_{\text{T1}},\\ &F^1_{\text{T2}},\ F^2_{\text{T2}},\ \ldots,\ F^L_{\text{T2}}] \end{aligned} \tag{8}$$

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- **TMB2**: Tokens of the four modalities are concatenated in reverse order:

$$\begin{aligned} F_{\text{TMB2}} = [&F^L_{\text{T2}},\ F^{L-1}_{\text{T2}},\ \ldots,\ F^1_{\text{T2}},\\ &F^L_{\text{T1}},\ F^{L-1}_{\text{T1}},\ \ldots,\ F^1_{\text{T1}},\\ &F^L_{\text{T1c}},\ F^{L-1}_{\text{T1c}},\ \ldots,\ F^1_{\text{T1c}},\\ &F^L_{\text{Flair}},\ F^{L-1}_{\text{Flair}},\ \ldots,\ F^1_{\text{Flair}}] \end{aligned} \tag{9}$$

- **TMB3**: Tokens are selected in an interleaved manner from the four modalities:

$$\begin{aligned} F_{\text{TMB3}} = [&F^1_{\text{Flair}},\ F^1_{\text{T1c}},\ F^1_{\text{T1}},\ F^1_{\text{T2}},\\ &F^2_{\text{Flair}},\ F^2_{\text{T1c}},\ F^2_{\text{T1}},\ F^2_{\text{T2}},\\ &\ldots,\\ &F^L_{\text{Flair}},\ F^L_{\text{T1c}},\ F^L_{\text{T1}},\ F^L_{\text{T2}}] \end{aligned} \tag{10}$$

- **TMB4**: Tokens are selected in an interleaved manner from the four modalities, and reversed:

$$\begin{aligned} F_{\text{TMB4}} = [&F^L_{\text{T2}},\ F^L_{\text{T1}},\ F^L_{\text{T1c}},\ F^L_{\text{Flair}},\\ &F^{L-1}_{\text{T2}},\ F^{L-1}_{\text{T1}},\ F^{L-1}_{\text{T1c}},\ F^{L-1}_{\text{Flair}},\\ &\ldots,\\ &F^1_{\text{T2}},\ F^1_{\text{T1}},\ F^1_{\text{T1c}},\ F^1_{\text{Flair}}] \end{aligned} \tag{11}$$

Each traversal extracts distinct output features from SSMs, which are then added to form the decoder's input. Note that, to ensure consistency, tokens must be reordered in a uniform manner before summation.

4) Shared Feature Decoder: Unlike traditional MIM approaches [41] that typically use an MLP for direct dimensional projection, we utilize a 3D convolutional decoder to reconstruct multimodal images. Specifically, we reshape the concatenated features produced by the multimodal traversals into a low-resolution 3D image (4D tensor with three spatial dimensions and one channel dimension) Z. A sequence of up-convolution blocks, each composed of two consecutive $3 \times 3 \times 3$ convolution layers and an upsampling layer, is then employed to progressively restore the spatial resolution from the high-dimensional latent space to the original image space. Denoting as D_{rec} the convolutional decoder used in the pre-training step, the predicted reconstruction can be expressed as

$$\hat{X} = D_{\text{rec}}(Z) \ \in \mathbb{R}^{|\mathcal{M}| \times H \times W \times D} \tag{12}$$

Let $\Omega = \{1, \ldots, H \times W \times D\}$ be the set of voxel indexes. The encoder and reconstruction decoder are trained without labeled data by minimizing the mean squared reconstruction error, as follows:

$$\mathcal{L}_{\text{rec}} = \frac{1}{|\mathcal{M}| \times |\Omega|} \sum_{m=1}^{|\mathcal{M}|} \sum_{i=1}^{|\Omega|} (\hat{x}_{m,i} - x_{m,i})^2 \tag{13}$$

where $\hat{x}_{m,i}$ is the reconstruction of modality m for voxel i.

C. Segmentation Fine-Tuning

The self-supervised pre-training step, outlined in Section III-B, leverages contextual and structural information to extract meaningful features, enabling effective learning of the downstream segmentation task with minimal annotated data. In this section, we detail how the pre-trained encoder is utilized to construct the final network for brain tumor image segmentation.

1) Segmentation Network: As shown in Fig. 4, several important changes are made to the architecture in the segmentation fine-tuning step. First, the reconstruction decoder is replaced by a segmentation decoder D_{seg} which is also composed of convolutional blocks. Following segmentation networks like 3D U-Net [43], we also introduce skip connections between the encoder and decoder to preserve fine details for segmentation.

In our architecture, each modality has a separate encoder, while a single decoder processes the combined features. This setup prevents the use of standard skip connections designed for symmetric encoder-decoder models. We solve this problem using tri-orientated spatial Mamba (TSMamba) blocks [18] in each layer encoder output feature fusion, which effectively integrate features from different modalities within each skip connection. A TSMamba block consists of a gated spatial convolution module (GSC), a tri-orientated Mamba module (ToM), and an MLP layer. The GSC module transforms this input using convolution blocks and a residual connection. The ToM module then tiles the 3D input features into three sequences, corresponding to the forward (z_f), reverse (z_r), and inter-slice (z_s) directions, and processes each of them with Mamba.

2) Auxiliary Regularizers: In brain tumor segmentation, some MRI sequences, such as FLAIR, provide more discriminative information than others. Consequently, the model may over-rely on these modalities, leading to significant accuracy degradation when they are unavailable. To mitigate this and promote balanced learning across all modalities, as shown in Fig. 4, we incorporate auxiliary regularizers inspired by those in [3].

To regularize separate convolutional encoders, we first upsample their outputs using a shared decoder having the same architecture as the convolutional decoder at the end of our network. We then encourage the segmentation for each modality m to match the ground-truth labels using a regularization loss $\mathcal{L}^m_{\text{reg}}$. Similarly, we regularize the decoder of our model by progressively upsampling the feature maps in each of its layer through deconvolution, and then obtaining dense predictions from the upsampled features with an additional softmax layer. The deep supervision loss for layer ℓ, denoted as $\mathcal{L}^\ell_{\text{DS}}$, measures the error between the ground truth segmentation and the prediction at this layer.

D. Loss Functions

The overall objective for fine-tuning the segmentation model combines the modality-specific regularization losses $\mathcal{L}^m_{\text{reg}}$, the deep supervision losses $\mathcal{L}^\ell_{\text{DS}}$, and the final segmentation loss $\mathcal{L}_{\text{seg}}$:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{seg}} + \lambda_{\text{reg}} \sum_{m=1}^{|\mathcal{M}|} \mathcal{L}^m_{\text{reg}} + \lambda_{\text{DS}} \sum_{\ell=1}^{L_{\text{dec}}} \mathcal{L}^\ell_{\text{DS}}. \tag{14}$$

Here, L_{dec} is the number of decoder layers and λ_{reg}, λ_{DS} are hyper-parameters controlling the relative weight of each loss term. For $\mathcal{L}^m_{\text{reg}}$, $\mathcal{L}^\ell_{\text{DS}}$, and $\mathcal{L}_{\text{seg}}$, we use a combination of weighted

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TABLE I
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON BRATS2018 DATASET

Type	Flair / T1 / T1c / T2	○ ○ ○ ●	○ ○ ● ○	○ ○ ● ○	● ○ ○ ○	○ ○ ● ●	○ ○ ● ○	● ● ○ ○	○ ● ○ ○	● ○ ● ○	● ○ ● ○	● ● ○ ○	● ● ○ ○	● ● ○ ○	○ ● ● ●	● ● ○ ●	Avg.
WT	RFNet	79.08	**67.99**	68.41	78.83	81.24	73.12	82.80	81.50	84.49	81.40	83.05	85.08	84.72	82.18	85.17	79.94
	mmFormer	78.99	67.56	66.97	78.82	80.19	72.60	81.76	81.02	83.67	81.40	82.83	84.51	84.69	81.22	84.86	79.41
	SRMNet	78.91	65.95	66.53	79.38	81.09	72.92	82.73	81.39	84.12	81.90	83.53	84.89	84.82	82.40	85.37	79.73
	M²FTrans	78.98	67.36	68.55	80.48	80.59	72.19	81.45	81.45	83.88	81.85	82.32	84.15	83.73	81.64	83.83	79.50
	Sup. Baseline (ours)	79.10	66.70	68.33	**80.73**	81.38	72.54	83.31	81.58	85.20	82.34	83.38	85.22	84.86	82.45	85.16	80.15
	M²SegMamba (ours)	**80.13**	67.80	**71.52**	80.66	**81.77**	**74.51**	**84.33**	**82.84**	**85.28**	**82.63**	**84.13**	**86.04**	**85.20**	**83.52**	**85.8**	**81.08**
TC	RFNet	55.53	59.30	48.08	54.27	67.53	63.67	59.29	58.23	61.95	63.43	67.33	62.75	69.75	68.94	70.49	62.04
	mmFormer	55.50	58.56	47.37	51.70	65.21	61.86	56.39	58.32	59.75	62.75	65.33	61.10	68.17	67.08	68.55	60.51
	SRMNet	53.81	58.62	42.56	51.91	68.41	63.96	56.3	55.48	59.43	65.8	67.64	60.13	69.92	69.07	70.36	60.89
	M²FTrans	**57.96**	60.13	51.03	55.11	64.59	61.83	57.81	**60.62**	62.33	63.17	63.86	63.22	65.74	65.39	66.49	61.29
	Sup. Baseline (ours)	57.25	60.83	48.05	**55.45**	68.31	65.27	60.22	60.24	**62.44**	67.13	68.91	63.64	69.78	69.60	70.86	63.20
	M²SegMamba (ours)	56.10	**62.34**	**52.75**	55.14	**69.28**	**67.48**	**61.69**	59.79	62.26	**68.38**	**70.61**	**63.83**	**71.64**	**71.40**	**72.46**	**64.34**
ET	RFNet	26.50	54.27	17.85	25.86	57.13	56.65	29.41	28.65	30.43	57.39	57.61	31.99	57.70	57.36	57.51	43.08
	mmFormer	26.97	53.46	20.65	25.25	55.91	53.95	27.89	28.56	29.37	55.60	55.38	30.62	55.57	55.54	54.90	41.97
	SRMNet	24.43	54.95	16.97	24.16	57.28	56.34	26.82	25.86	28.64	56.79	56.19	29.14	56.07	56.50	55.45	41.71
	M²FTrans	26.89	54.43	16.82	**27.64**	54.83	54.82	28.95	28.28	30.32	55.90	56.12	30.90	55.66	55.25	55.76	42.17
	Sup. Baseline (ours)	26.92	**54.99**	18.87	25.73	57.72	56.83	28.97	28.86	30.58	**57.58**	57.13	32.12	57.03	57.01	56.62	43.13
	M²SegMamba (ours)	**27.80**	54.61	**24.29**	25.77	**58.31**	**57.34**	**30.18**	**30.07**	**30.74**	57.12	**57.88**	**32.54**	**57.82**	**58.43**	**57.78**	**44.05**

Here, Supervised Baseline (ours) is the segmentation result without pre-training parameters, and M²SegMamba (ours) is the fine-tuned segmentation result with pre-training parameters.

cross-entropy (WCE) and Dice loss to measure the loss for each sample in the batch.

$$\mathcal{L} = \mathcal{L}_{\text{WCE}} + \mathcal{L}_{\text{Dice}} \tag{15}$$

Denoting as $Y, \hat{Y} \in [0,1]^{|\Omega| \times K}$ the ground-truth and predicted segmentation for image X, where K is the number of segmentation classes, these losses can be expressed as

$$\mathcal{L}_{\text{WCE}} = -\sum_{k=1}^{K} w_k \sum_{i=1}^{|\Omega|} y_{ik} \log(\hat{y}_{ik}) \tag{16}$$

$$w_k = 1 - \frac{\sum_{i=1}^{|\Omega|} y_{ik}}{\sum_{k=1}^{K} \sum_{i=1}^{|\Omega|} y_{ik}} \tag{17}$$

$$\mathcal{L}_{\text{Dice}} = 1 - \frac{1}{K} \sum_{k=1}^{K} \frac{2 \sum_{i=1}^{|\Omega|} y_{ik} \hat{y}_{ik}}{\sum_{i=1}^{|\Omega|} y_{ik} + \sum_{i=1}^{|\Omega|} \hat{y}_{ik} + \epsilon}, \tag{18}$$

where w_k denotes the weight of the kth class and $\epsilon = 10^{-7}$ is used to avoid division by zero.

IV. EXPERIMENTS AND RESULTS

A. Dataset and Implementation

1) Dataset: The experiments were conducted using two datasets of the multimodal Brain Tumor Segmentation (BraTS) challenge [44]: BraTS 2018 and BraTS 2020. These datasets comprise four MRI modalities: Flair, T1c, T1, and T2. The BraTS 2018 dataset includes 190 training cases and 95 validation cases, while the BraTS 2020 dataset contains 246 training cases and 123 validation cases. To simulate a few-shot learning scenario, the segmentation model training was performed using 10% of the available training data (results for different training data ratios are reported in Table V). All images from each modality were preprocessed by normalizing and resizing them to a uniform resolution of $128 \times 128 \times 128$. The performance of the proposed method was evaluated based on segmentation results for the whole tumor (WT), tumor core (TC), and enhancing tumor (ET) regions. The Dice similarity coefficient was employed as the primary metric to quantitatively assess segmentation accuracy.

2) Implementation Details: All images in the training set (100%) are used for the unsupervised pre-training step, ignoring the labels of these images. Optimization was performed with Adam with a learning rate of 1×10^{-4}, a batch size of 2, and a total of 300 training epochs. For fine-tuning, we used an initial learning rate of 2×10^{-4}, a weight decay of 1×10^{-4}, a batch size of 1, and a total of 1,000 training epochs.

For the main experiments, we set the mask ratio empirically to $\gamma = 87.5\%$. The impact of this ratio is later investigated in Section IV-C2.

B. Comparison With State-of-The-Art

In this section, we compare our method with state-of-the-art brain tumor segmentation models, including RFNet [5], mmFormer [3], SRMNet [8], and M²FTrans [9]. Tables I and II present the quantitative comparison results across 15 different multimodal combinations, for the BraTS 2018 and BraTS 2020 datasets. The best and second-best results for each modality setting are highlighted in bold and underlined, respectively.

The proposed models (Supervised Baseline without pre-training, as well as our M²SegMamba model with pre-training and fine-tuning), demonstrate outstanding performance across all combinations. On the BraTS 2018 dataset, RFNet, which utilizes CNNs to implement a region-aware fusion module, achieves strong results. Compared to RFNet, our method yields average improvements of 1.14%, 2.30%, and 0.97% for WT, TC, and ET, respectively. Similarly, when compared to the baseline method mmFormer, the average gains are 1.67%, 3.83%, and

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TABLE II
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON BRATS2020 DATASET

Type	Flair	○	○	○	●	○	○	●	○	●	●	●	●	●	○	●	Avg.
	T1	○	○	●	○	○	●	●	●	○	○	●	●	○	●	●	
	T1c	○	●	○	○	●	●	○	○	○	●	●	○	●	●	●	
	T2	●	○	○	○	●	○	○	●	●	○	○	●	●	●	●	
WT	RFNet	80.38	70.52	69.56	82.91	82.13	74.94	85.80	82.54	87.59	85.72	86.44	88.01	87.81	82.88	87.87	82.34
	mmFormer	80.82	68.90	67.82	82.40	82.83	72.83	85.10	82.26	86.90	84.80	85.68	87.44	87.35	82.79	87.48	81.69
	SRMNet	79.77	68.70	65.67	83.98	82.20	72.52	85.25	81.23	87.33	85.32	85.93	87.33	86.94	82.73	87.24	81.48
	M^2FTrans	79.66	70.99	69.02	84.26	82.41	74.00	85.57	81.57	86.58	86.64	86.51	87.34	87.94	82.63	87.88	82.20
	Sup. Baseline (ours)	**80.96**	70.12	67.31	83.51	83.37	74.45	85.35	82.78	87.78	86.71	86.95	87.91	88.39	83.86	88.31	82.52
	M^2SegMamba (ours)	80.01	**72.43**	**71.03**	**84.43**	**83.88**	**76.62**	**87.32**	**83.49**	**88.63**	**87.06**	**87.93**	**89.24**	**88.93**	**84.73**	**89.23**	**83.66**
TC	RFNet	49.78	72.16	46.13	53.11	75.13	73.53	58.48	55.16	58.24	75.02	75.97	59.90	76.12	75.20	76.07	65.33
	mmFormer	55.22	70.43	46.12	51.89	74.91	72.79	58.41	56.60	58.19	73.27	75.16	60.61	75.08	75.51	75.78	65.33
	SRMNet	50.02	72.40	41.08	53.51	75.07	73.55	56.81	52.48	57.89	74.34	76.17	58.14	75.15	76.18	76.21	64.60
	M^2FTrans	56.08	71.72	48.75	53.84	74.60	73.45	57.95	60.53	59.33	75.13	75.77	61.79	76.25	75.33	76.61	66.48
	Sup. Baseline (ours)	**57.70**	72.75	44.41	52.69	77.69	74.86	58.49	60.93	61.65	77.68	78.23	62.97	79.57	78.49	79.84	67.86
	M^2SegMamba (ours)	57.15	**74.64**	**50.02**	**55.65**	**78.88**	**78.07**	**62.92**	**61.83**	**63.27**	**79.36**	**81.03**	**65.22**	**80.62**	**80.47**	**81.29**	**70.04**
ET	RFNet	25.27	64.82	20.19	30.36	66.28	66.24	31.44	28.73	31.54	67.71	67.70	32.70	67.09	65.83	66.71	48.84
	mmFormer	30.53	63.41	20.61	28.84	66.51	65.73	31.22	31.60	32.65	65.49	67.26	33.86	65.92	66.74	66.07	49.10
	SRMNet	26.53	63.25	18.24	29.98	66.89	64.83	30.62	28.78	33.06	66.21	67.47	32.21	66.36	66.94	66.72	48.54
	M^2FTrans	31.50	63.16	**22.80**	29.45	65.52	64.85	31.48	34.34	33.74	65.95	66.40	34.86	66.10	65.72	66.12	49.47
	Sup. Baseline (ours)	**32.94**	64.05	18.20	30.20	68.62	67.83	31.11	32.66	34.96	67.70	68.87	34.83	67.79	68.39	68.14	50.47
	M^2SegMamba (ours)	32.80	**65.02**	21.93	**30.84**	**68.96**	**69.15**	**34.22**	**34.73**	**35.50**	**68.84**	**70.21**	**36.81**	**68.62**	**70.05**	**69.92**	**51.84**

Here, Supervised Baseline (ours) is the segmentation result without pre-training parameters, and M^2SegMamba (ours) is the fine-tuned segmentation result with pre-training parameters.

2.08%. M^2FTrans, a more recent self-attention-based approach incorporating learnable fusion tokens and masked self-attention, is also outperformed by our method, with average improvements of 1.58%, 3.05%, and 2.33%.

The segmentation results on the BraTS 2020 dataset also demonstrate consistent improvements, with average increases of 1.32%, 4.71%, and 3.00% compared to RFNet, and 1.97%, 4.71%, and 2.74% compared to mmFormer. These results validate the stability of our proposed method across various modality settings. Furthermore, fine-tuning after pre-training yields additional gains over the Supervised Baseline, with improvements of 0.93%, 1.14%, and 0.92% in WT, TC, and ET for BraTS 2018, and 1.14%, 2.18%, and 1.37% for BraTS 2020. This performance boost is primarily attributed to the self-supervised pre-training mask strategy, which effectively enhances feature representation learning for brain tumor images.

Figs. 5 and 6 present the qualitative comparison results of various modality combinations—including one, two, three, and all modalities—for the BraTS 2018 and BraTS 2020 datasets. These results underscore the benefits of incorporating multiple modalities into segmentation models, and demonstrate that our method consistently produces visually superior segmentations across all modality combinations compared to other approaches. Although all methods effectively segment brain tumors, noticeable differences exist. For the BraTS 2018 dataset, all approaches achieve relatively accurate segmentation of the peritumoral edema (ED), but our proposed method demonstrates superior performance in segmenting the tumor core. As the number of modalities increases, the segmentation accuracy of our method across all tumor regions closely aligns with the ground truth. Similarly, for the BraTS 2020 dataset, our method maintains strong segmentation performance. The red detail box in the lower-left corner magnifies the blue box region, further highlighting our method's superior accuracy in segmenting the ED region.

TABLE III
ABLATION STUDY ON THE MAIN COMPONENTS OF OUR NETWORK

Components			Avg. DSC (%)		
Mamba	Multi-traversal	TSMamba	WT	TC	ET
×	×	×	79.41	60.51	41.97
✓	×	×	79.46	61.46	42.25
✓	✓	×	79.87	62.90	42.63
✓	✓	✓	**80.15**	**63.20**	**43.13**

C. Ablation Study

We conducted ablation experiments on the BraTS 2018 dataset to evaluate the impact of the proposed model's individual components, the mask ratio employed in pre-training, the amount of labeled data for segmentation fine-tuning, and our model's computational efficiency for increasing input sizes.

1) Network Components: The effectiveness of each component in our network is validated in Table III. Specifically, we used mmFormer as base architecture and modified it by replacing the ViT blocks of its encoder by our Mamba blocks, adding the multimodal traversal, and/or incorporating skip connections via TSMamba blocks. As can be seen, using Mamba blocks in the encoder improves the segmentation of all tumor regions, with a highest DSC improvement of 0.95% obtained for TC. Adding the proposed multimodal traversal further boosts performance by 0.74% on average for the three tumor regions. This highlights the traversal's ability to effectively extract and integrate feature information from multiple modalities, resulting in more robust and accurate predictions. Moreover, incorporating skip connections using TSMamba blocks strengthens the connection between the network's encoder and decoder, resulting in another 1.10% DSC increase averaged over all tumor regions. Lastly, the combination of these three elements achieves optimal segmentation performance, improving WT, TC, and

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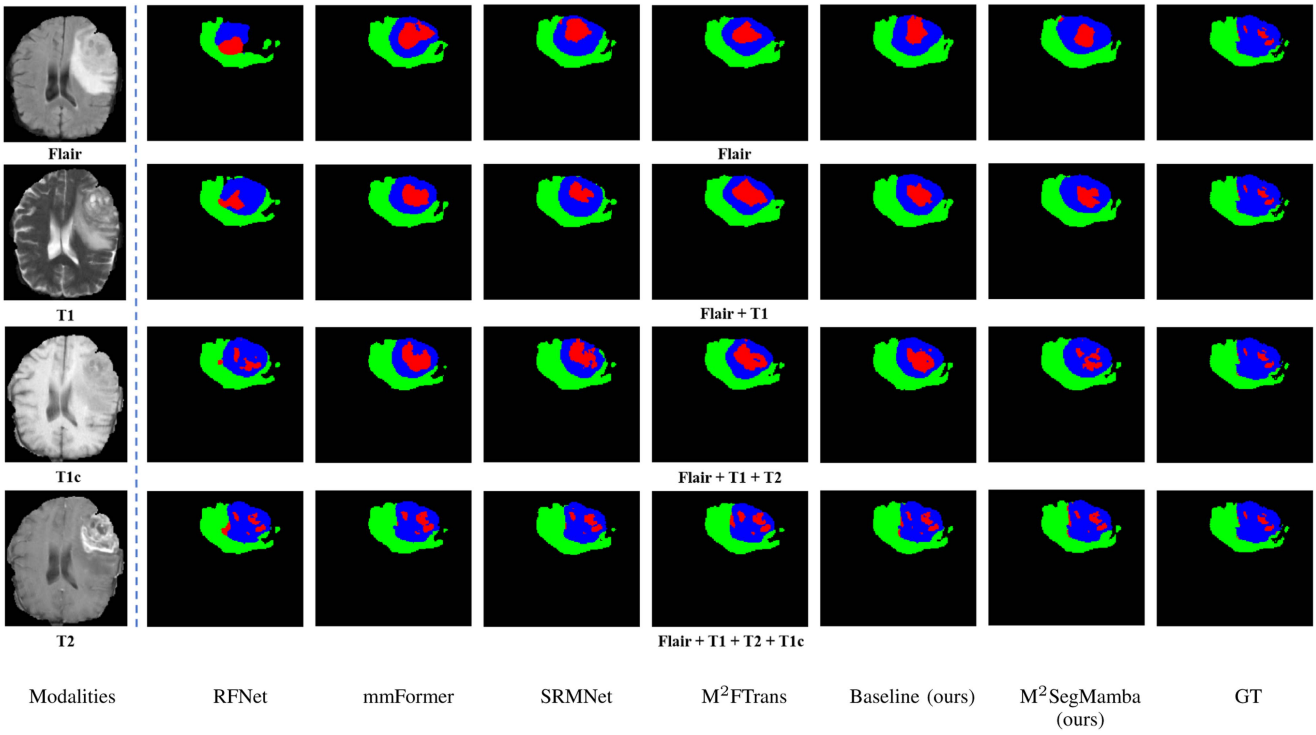


Fig. 5. Comparison of segmentation results of the BraTS 2018 dataset between our method and existing methods with different numbers of missing modalities. Here, Baseline (ours) is the segmentation result without pre-training parameters, and M^{2}SegMamba (ours) is the fine-tuned segmentation result with pre-training parameters.

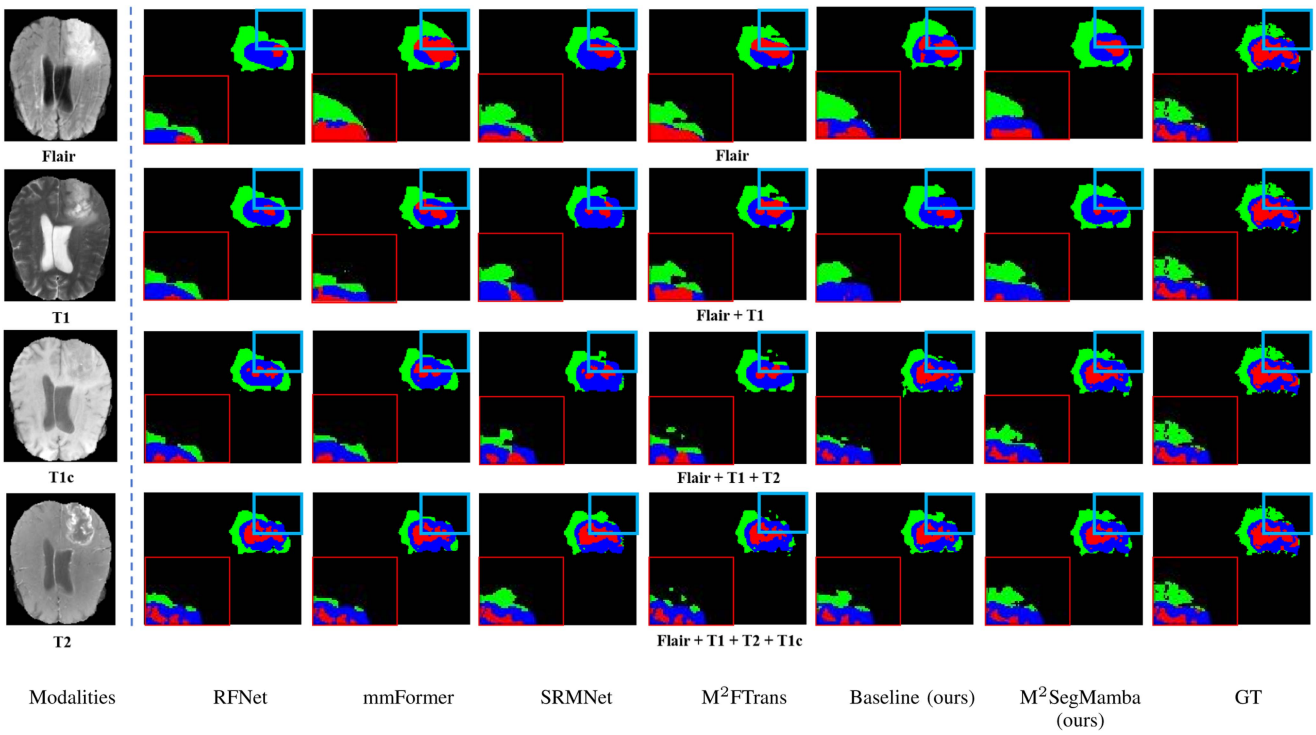


Fig. 6. Comparison of segmentation results of the BraTS 2020 dataset between our method and existing methods with different numbers of missing modalities.

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TABLE IV
RESULTS FOR DIFFERENT MASK RATIOS USED IN PRE-TRAINING

Masking Ratio	WT	TC	ET
87.5 %	**81.08**	64.34	**44.05**
75.0 %	80.36	**64.40**	43.84
50.0 %	80.47	64.05	43.52
25.0 %	80.41	63.18	43.26

TABLE V
PERFORMANCE COMPARISON ON DIFFERENT TRAINING RATIOS USING THE BRATS2018 DATASETS

Method	Training ratio	WT	TC	ET
mmFormer	10 %	79.41	60.51	41.97
M^3AE	10 %	78.22	64.02	41.82
Supervised Baseline (ours)	10 %	80.15	63.20	43.13
M^2SegMamba (ours)	10 %	**81.08**	**64.34**	**44.05**
mmFormer	50 %	85.03	73.20	49.85
M^3AE	50 %	84.87	75.84	50.41
Supervised Baseline (ours)	50 %	86.32	76.13	52.22
M^2SegMamba (ours)	50 %	**86.73**	**76.81**	**53.01**
mmFormer	100 %	86.64	76.89	52.79
M^3AE	100 %	87.12	75.10	**54.70**
Supervised Baseline (ours)	100 %	87.27	77.99	53.90
M^2SegMamba (ours)	100 %	**87.46**	**78.07**	54.23

ET by 0.74%, 2.69%, and 1.16% compared to the baseline, respectively.

2) Mask Ratio in Pre-Training: We conducted experiments using different mask ratios in self-supervised pre-training to investigate the impact of this hyperparameter on segmentation performance. As shown in Table IV, our masking strategy improves segmentation results across all mask ratios. In general, using a higher mask ratio enhances downstream performance by compelling the network to learn both intra- and inter-modality dependencies more effectively during pre-training.

3) Amount of Labeled Data for Fine-Tuning: To thoroughly evaluate our model's generalization capability, we conducted experiments using varying proportions of the training set—10%, 50%, and 100%—and compared both pretrained and non-pretrained versions of our method against the baseline model (mmFormer). In addition, to improve the fairness and completeness of the comparison, we incorporated M^3AE, a representative pretrain-finetune method for multi-modal medical image modeling, as an additional baseline. This allows for a more comprehensive evaluation of our self-supervised pretraining strategy under comparable conditions.

Table V presents the comparison results, demonstrating that our method consistently outperforms others in segmentation across various training data ratios. Notably, for the self-supervised pretrained model, overall segmentation performance improves, with the most significant gains observed on small size datasets. This highlights the effectiveness of our proposed pretraining approach for few shot segmentation tasks.

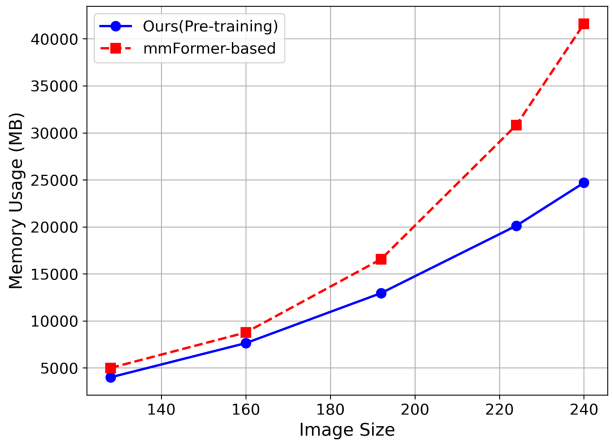


Fig. 7. Model efficiency comparison of pre-training.

4) Model Efficiency: To assess the efficiency of our Mamba-based method, for increasing input image sizes, we compared it to the mmFormer baseline, which utilizes ViT blocks. As shown in Fig. 7, our method's memory usage grows significantly slower than mmFormer, reducing memory consumption by over 40% for the largest image size of 240. By leveraging the Mamba-based architecture, our model minimizes redundant computations and optimizes resource allocation, making it especially well-suited for high-resolution medical image processing tasks

V. DISCUSSION AND CONCLUSION

To address the problems of missing modalities and few-shot scenarios in clinical practice, we propose a novel Mamba-based framework named M^2SegMamba for few-shot and incomplete multimodal brain tumor image segmentation. Our approach innovatively integrates Masked Image Modeling with 3D multimodal brain tumor segmentation, showing great potential in small-sample scenarios. Leveraging the advantages of Mamba for long-sequence modeling, we design Mamba-based network, which further enhance the segmentation accuracy of the network. Compared to Transformer-based methods, our approach maintains excellent inference efficiency. Experiments on the BraTS 2018 and BraTS 2020 datasets validate the effectiveness of our proposed method. Clinically, M^2SegMamba's ability to perform with limited data and incomplete modalities highlights its potential as a practical tool for real-world applications, especially in resource-constrained healthcare environments.

While our proposed M^2SegMamba framework has demonstrated robust performance in the challenging settings of few-shot and incomplete multimodal segmentation tasks, the model still exhibits certain performance variations under specific conditions, highlighting areas for potential improvement. For instance, segmentation accuracy may vary across different tumor subregions when critical imaging modalities are absent. These differences likely stem from the inherent heterogeneity of tumor structures and the diverse information content across imaging modalities. In addition, while this study focuses on MRI-based segmentation, the modular architecture and adaptive multimodal traversal strategy of M^2SegMamba indicate its potential for

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extension to other imaging modalities. Future work will focus on enhancing model robustness in low-information scenarios and exploring its applicability to broader imaging domains.

In summary, the current study lays a promising foundation, but further investigations are required to fully unlock the versatility and scalability of the framework across diverse medical imaging scenarios. Building upon the encouraging results presented here, future work will focus on extending the applicability of this framework beyond its current scope, addressing a broader range of real-world clinical challenges and datasets.

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